

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460220.9081 +/- 0.0058 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1486 +/- 0.0098
Transit depth (Rp/Rs)^2	2.21 +/- 0.29 [%]
Orbital Inclination (inc)	88.55 +/- 1.21
Airmass coefficient 1 (a1)	1.1327 +/- 0.008
Airmass coefficient 2 (a2)	-0.096 +/- 0.015
Scatter in the residuals of the lightcurve fit is	0.71 %
Best Comparison Star	None
Optimal Aperture	3.47
Optimal Annulus	11.2
Transit Duration (day)	0.1155 +/- 0.0067

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1486 ± 0.0098 vs 0.1276 (deviation 2.14σ)
Duration fit vs published	0.1155 d vs 0.1179 d (deviation 0.36σ)
Mid-time O–C	-33.2 min
Depth SNR	15.2
Residual scatter	0.71 %

Light curve

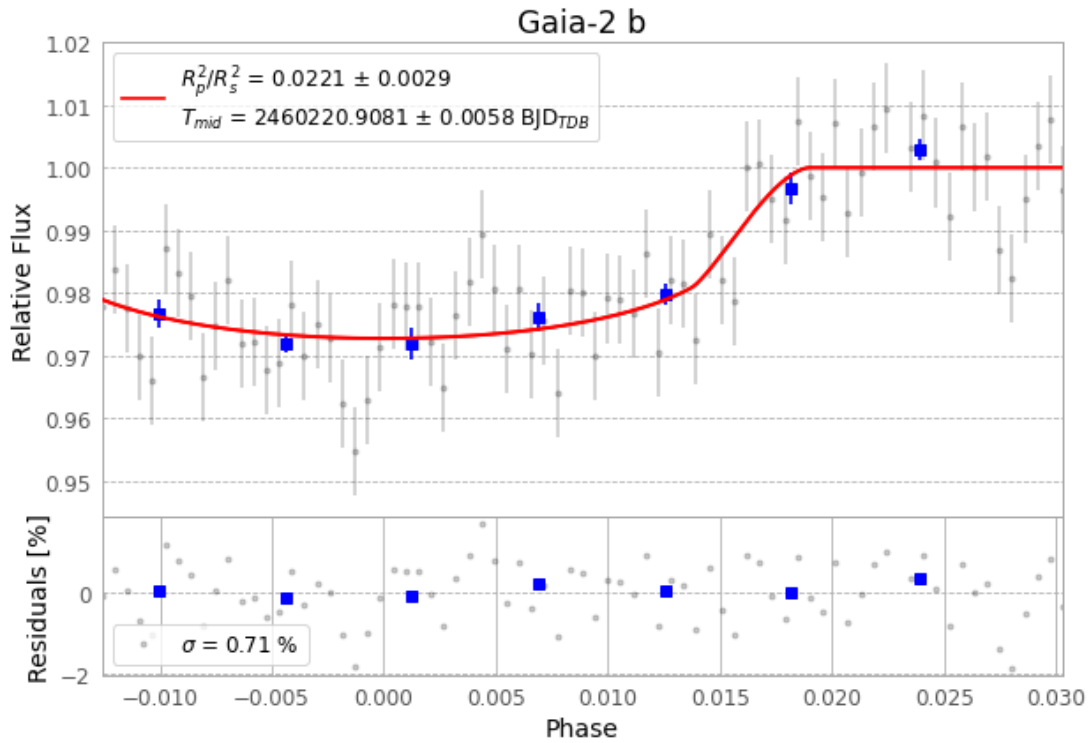


Figure 1 — FinalLightCurve_Gaia-2 b_2023-10-03.png (SHA-256 ea759d6b07fa11e6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [326, 261], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a512ea9cd27d995c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

77 FITS frames (51 MB), Gaia-2231003083915.FITS → Gaia-2231003122716.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Gaia-2-231003.log (a740d1a30658...), FinalLightCurve_Gaia-2 b_2023-10-03.png (ea759d6b07fa...), FinalLightCurve_Gaia-2 b_2023-10-03.csv (d7070b66d9a3...)

Compute resources

Wall time 113.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-18 01:16Z.